

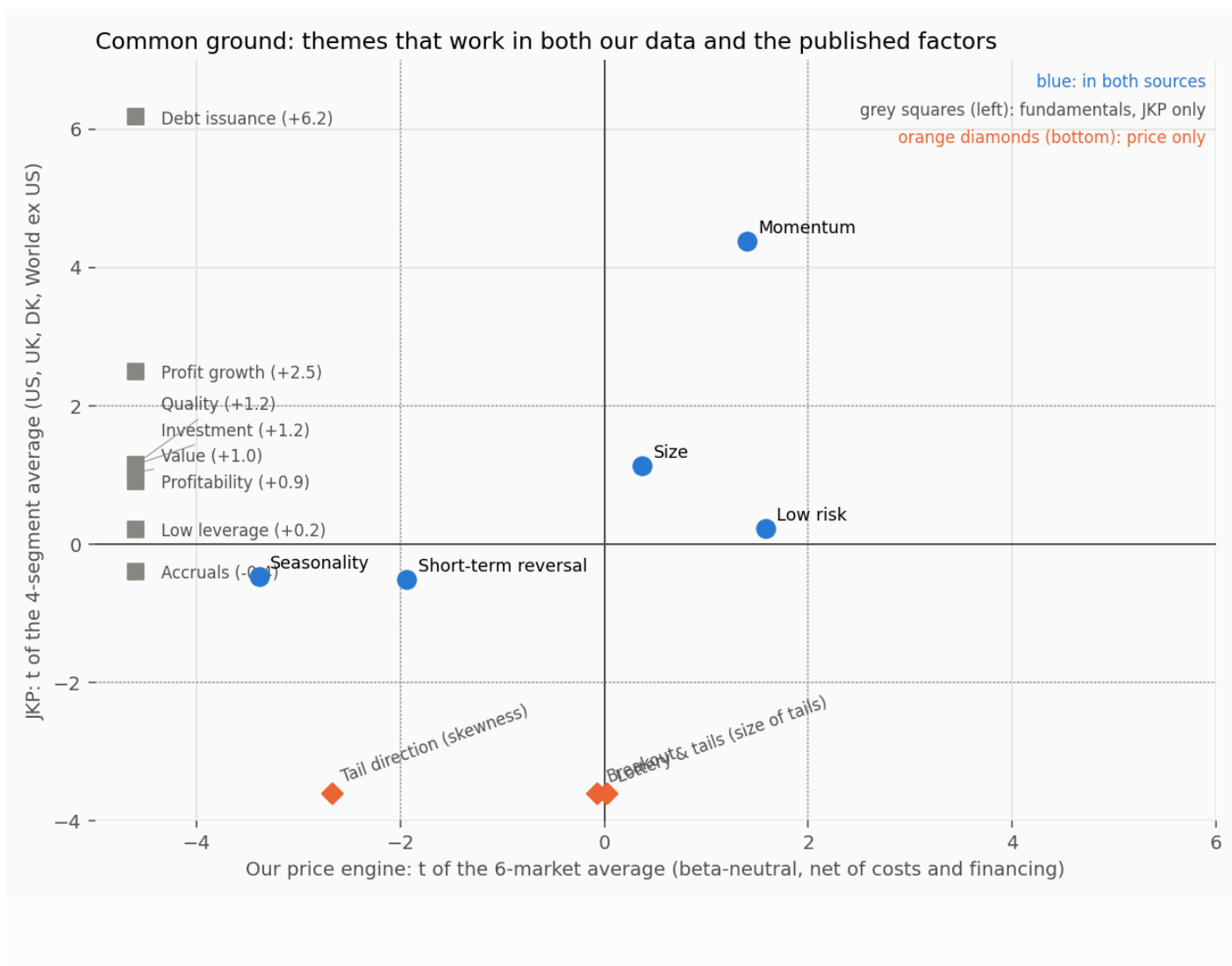
FACTSHEET FS13 · The signal battery

Every signal in every market: where is the common ground?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · price data to 31 August 2026, JKP to December 2025

Key facts	
Price battery	19 price signals × 6 point-in-time universes (US, EU, UK, DK, SCANDI, World), monthly, 2013–2026, our engine, net of 10 bp per side and a size-tiered borrow fee on the short leg
Published battery	153 JKP factors (Jensen, Kelly & Pedersen 2023) × 5 segments (US, UK, Denmark, World ex US, World), 2013–2025, grouped into 13 themes
Direction	fixed from the literature before the run (table in §1); no signal was flipped after seeing results
Books	long/short (top minus bottom quantile), beta-neutral long/short (charged for the cash it borrows at the USD risk-free rate + 50 bp), long-only vs the equal-weight universe
Pooled test	equal-weight average of the markets; its t-statistic includes the co-movement between markets
Multiple testing	Benjamini–Hochberg at 5% over 38 price tests (19 signals × raw/beta-neutral) and over 153 JKP factors; Harvey–Liu–Zhu $t > 3$ as a second bar

In one paragraph. No price signal earns a positive return that survives the multiple-testing correction once short legs pay borrow fees and the beta-neutral books pay for the leverage they use; thirteen years of large caps are too short for more. What does survive is a loser: Same-month seasonality (beta-neutral) reliably loses money. The common ground is in the *direction*. **Low beta** is positive in 6 of 6 of our markets after all costs (t 2.4) but flat in JKP's value-weighted factors. **Momentum** is positive in 5 of 6 of our markets (12-1 momentum 5 of 6, residual momentum 6 of 6) and in all 4 JKP segments (t 4.4); in our large caps it is not significant on its own (t 1.4). Among the fundamental themes we cannot yet build ourselves, **debt issuance** (t 6.2) and **profit growth** (t 2.5) worked in every JKP segment since 2013. **Same-month seasonality** and **tail direction** are reliably negative in our large caps. The 19 price signals hold only a handful of independent bets: one low-risk block of nine and one momentum block of four, the same structure in the US, EU and World (adjusted Rand index 0.87–0.97). SCANDI and UK agree least with the other markets on which signals work. The engine reproduces the published US factors (correlation ≥ 0.8 for 8 of 12 matched signals).



1. What was tested

Two batteries, one question: which signals work across segments, and which only in one corner?

Price battery (our data, our engine). Every signal in the library (`code/signals.py`, `signals2.py`) runs through the same sort engine as FS03–FS12 (`code/xs.py`) in all six universes: monthly quantile sort (deciles ≥ 100 names, quintiles 50–99, terciles below), equal weight, 10 bp per side on turnover. Three books per signal: long/short, **beta-neutral** long/short (each leg scaled to beta 1 before netting, so the market bet that dominated FS02 and FS04 is removed), and long-only against owning all stocks equally. The direction of every signal was set from the literature before running:

Signal	Family	Long the	Direction from
Low MAX (1 day)	Lottery & tails	low	Bali, Cakici & Whitelaw 2011
Low MAX (5 days)	Lottery & tails	low	Bali, Cakici & Whitelaw 2011
Mild worst day (MIN)	Lottery & tails	high	low-risk convention: small crashes
Mild worst 5 days	Lottery & tails	high	low-risk convention
Narrow daily range	Lottery & tails	low	low-risk convention
Low net tail	Lottery & tails	low	Boyer, Mitton & Vorkink 2010 (low skew)
Low skewness	Lottery & tails	low	Boyer, Mitton & Vorkink 2010

Signal	Family	Long the	Direction from
Low idiosyncratic vol	Low risk	low	Ang, Hodrick, Xing & Zhang 2006
Low volatility (63d)	Low risk	low	Blitz & van Vliet 2007
Low volatility (252d)	Low risk	low	Blitz & van Vliet 2007
Low beta	Low risk	low	Frazzini & Pedersen 2014
Short-term reversal	Reversal	low	Jegadeesh 1990
Momentum 12-1	Momentum	high	Jegadeesh & Titman 1993
Residual momentum	Momentum	high	Blitz, Huij & Martens 2011
Same-month seasonality	Seasonality	high	Heston & Sadka 2008
Near 52-week high	Momentum	high	George & Hwang 2004
Far above 52-week low	Momentum	high	momentum-consistent (falling knife loses)
Near 55-day high	Momentum	high	Turtle breakout, cross-sectional
Small size	Size	low	Banz 1981

Two directions are conventions rather than paper results: MIN and the daily range follow the low-risk reading (small tails are good), and "far above the 52-week low" is the momentum reading, which FS12 had already shown. Treat those three as less than fully out-of-sample.

Published battery (JKP). The 153 factors of Jensen, Kelly & Pedersen (2023, *Is There a Replication Crisis in Finance?*), capped value-weighted long/short, in USD, signed so the original paper's prediction is positive, grouped into their 13 themes. JKP covers fundamentals (value, quality, investment, profitability, accruals, debt issuance) that we do not yet have outside the US. Segments: US, UK, Denmark, World ex US (disjoint from the US) and World. **There is no JKP file for EU or Scandinavia in our cache and the JKP site is not reachable from the build environment; World ex US stands in for both where a comparison is needed.**

2. The price battery: every signal, every market

19 price signals × 6 markets, 2013–2026 (rows ordered by return similarity)

Long/short Sharpe, net

Small size	+0.04	-0.18	+0.45	-0.21	+0.25	+0.02
Short-term reversal	-0.36	-0.61	-0.17	-0.38	+0.01	-0.49
Low net tail	-0.67	-0.19	-0.51	-0.21	+0.17	-0.58
Low skewness	-0.92	-0.12	-0.73	-0.46	+0.05	-0.90
Same-month seasonality	-0.67	-0.83	+0.22	-0.85	-0.60	-0.30
Low beta	-0.54	-0.16	+0.01	-0.34	-0.37	-0.41
Low volatility (63d)	-0.51	-0.15	-0.12	-0.01	-0.25	-0.38
Low volatility (252d)	-0.60	-0.08	-0.30	-0.08	-0.26	-0.48
Narrow daily range	-0.69	-0.06	-0.18	-0.05	-0.09	-0.62
Mild worst day (MIN)	-0.65	-0.28	-0.12	-0.08	-0.50	-0.56
Mild worst 5 days	-0.50	-0.23	-0.07	-0.08	-0.55	-0.46
Low idiosyncratic vol	-0.64	+0.20	-0.18	+0.03	+0.22	-0.42
Low MAX (1 day)	-0.81	-0.09	-0.20	-0.26	-0.17	-0.69
Low MAX (5 days)	-0.72	-0.24	-0.19	-0.21	+0.01	-0.65
Residual momentum	+0.20	+0.25	+0.14	+0.46	+0.18	+0.18
Near 52-week high	-0.28	+0.20	-0.03	+0.33	-0.10	-0.20
Near 55-day high	-0.42	-0.14	-0.17	+0.30	-0.23	-0.36
Momentum 12-1	+0.32	+0.34	+0.43	+0.31	-0.19	+0.41
Far above 52-week low	+0.60	+0.60	+0.33	+0.19	+0.07	+0.56
	US	EU	UK	DK	SCANDI	World

Beta-neutral L/S Sharpe, after costs and financing

Small size	-0.32	-0.08	+0.66	-0.16	+0.23	-0.02
Short-term reversal	-0.52	-0.65	-0.32	-0.40	+0.02	-0.69
Low net tail	-0.70	-0.22	-0.53	-0.31	+0.11	-0.60
Low skewness	-1.00	-0.11	-0.59	-0.53	+0.00	-0.88
Same-month seasonality	-0.77	-0.93	+0.10	-0.91	-0.71	-0.32
Low beta	+0.13	+0.56	+0.98	+0.08	+0.09	+0.31
Low volatility (63d)	-0.03	+0.30	+0.45	+0.32	+0.02	+0.11
Low volatility (252d)	-0.07	+0.52	+0.29	+0.35	+0.09	+0.15
Narrow daily range	-0.37	+0.35	+0.27	+0.18	+0.13	-0.25
Mild worst day (MIN)	-0.37	+0.00	+0.32	+0.24	-0.31	-0.19
Mild worst 5 days	-0.23	+0.06	+0.49	+0.26	-0.37	-0.12
Low idiosyncratic vol	-0.36	+0.40	+0.08	+0.20	+0.20	-0.09
Low MAX (1 day)	-0.48	+0.30	+0.35	-0.09	-0.01	-0.30
Low MAX (5 days)	-0.36	+0.18	+0.40	+0.02	+0.29	-0.27
Residual momentum	+0.23	+0.25	+0.27	+0.37	+0.04	+0.19
Near 52-week high	-0.18	+0.40	+0.28	+0.48	+0.02	-0.04
Near 55-day high	-0.33	+0.05	+0.01	+0.47	-0.17	-0.24
Momentum 12-1	+0.29	+0.55	+0.70	+0.26	-0.10	+0.55
Far above 52-week low	+0.19	+0.31	+0.08	+0.02	-0.07	+0.14
	US	EU	UK	DK	SCANDI	World

Signal	Family	L/S net, 6-market average	L/S > 0	Beta-neutral, net of costs and financing (✓ = BH)	Beta-neutral > 0	Long-only beats EW Sharpe	Effective markets	Cost / yr	Beta-neutral Sharpe 2013–19 / 2020–26
Low beta	Low risk	-6.8% (t -1.5)	1 / 6	+10.0% (t +2.4)	6 / 6	4 / 6	1.7	+0.5%	+1.21 / +0.10
Momentum 12-1	Momentum	+5.6% (t +1.5)	5 / 6	+6.2% (t +2.0)	5 / 6	5 / 6	1.8	+1.2%	+0.66 / +0.40
Residual momentum	Momentum	+3.3% (t +1.2)	6 / 6	+3.0% (t +1.2)	6 / 6	6 / 6	2.1	+2.0%	+0.12 / +0.50
Low volatility (252d)	Low risk	-6.4% (t -1.4)	0 / 6	+3.5% (t +1.1)	5 / 6	5 / 6	1.8	+0.5%	+0.52 / +0.11
Low volatility (63d)	Low risk	-5.0% (t -1.1)	0 / 6	+3.1% (t +1.0)	5 / 6	6 / 6	1.8	+1.2%	+0.63 / +0.02
Near 52-week high	Momentum	-0.4% (t -0.1)	2 / 6	+2.7% (t +0.8)	4 / 6	5 / 6	1.6	+2.0%	+0.30 / +0.15
Far above 52-week low	Momentum	+6.4% (t +2.0)	6 / 6	+1.8% (t +0.6)	5 / 6	5 / 6	1.8	+1.6%	+0.17 / +0.15
Low idiosyncratic vol	Low risk	-2.3% (t -0.8)	3 / 6	+1.0% (t +0.4)	4 / 6	4 / 6	2.2	+2.9%	+0.11 / +0.13

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Signal	Family	L/S net, 6-market average	L/S > 0	Beta-neutral, net of costs and financing (✓ = BH)	Beta-neutral > 0	Long-only beats EW Sharpe	Effective markets	Cost / yr	Beta-neutral Sharpe 2013–19 / 2020–26
Small size	Size	+0.5% (t +0.3)	4 / 6	+0.8% (t +0.4)	2 / 6	1 / 6	2.9	+0.4%	+1.43 / -0.66
Narrow daily range	Lottery & tails	-5.1% (t -1.4)	0 / 6	+0.6% (t +0.2)	4 / 6	3 / 6	2.0	+3.1%	+0.20 / -0.03
Low MAX (5 days)	Lottery & tails	-6.1% (t -1.6)	1 / 6	+0.4% (t +0.1)	4 / 6	2 / 6	2.0	+3.1%	+0.68 / -0.42
Mild worst 5 days	Lottery & tails	-5.9% (t -1.4)	0 / 6	+0.4% (t +0.1)	3 / 6	3 / 6	1.8	+3.0%	-0.17 / +0.18
Near 55-day high	Momentum	-3.0% (t -0.9)	1 / 6	-0.2% (t -0.1)	3 / 6	3 / 6	1.7	+2.9%	-0.24 / +0.14
Mild worst day (MIN)	Lottery & tails	-5.7% (t -1.7)	0 / 6	-0.6% (t -0.2)	3 / 6	2 / 6	2.0	+3.4%	-0.24 / +0.07
Low MAX (1 day)	Lottery & tails	-5.9% (t -1.8)	0 / 6	-0.6% (t -0.2)	2 / 6	2 / 6	2.1	+3.4%	+0.35 / -0.38
Short-term reversal	Reversal	-5.5% (t -1.6)	1 / 6	-5.9% (t -1.9)	1 / 6	0 / 6	2.0	+3.9%	+0.06 / -1.15
Low net tail	Lottery & tails	-3.6% (t -1.9)	1 / 6	-3.8% (t -2.1)	1 / 6	0 / 6	2.4	+4.0%	-0.07 / -1.03
Low skewness	Lottery & tails	-4.6% (t -2.4)	1 / 6	-4.8% (t -2.8)	1 / 6	0 / 6	2.7	+4.0%	-0.55 / -1.02
Same-month seasonality	Seasonality	-6.5% (t -2.7)	1 / 6	-7.3% (t -3.4) ✓	1 / 6	1 / 6	2.3	+3.9%	-1.09 / -0.85

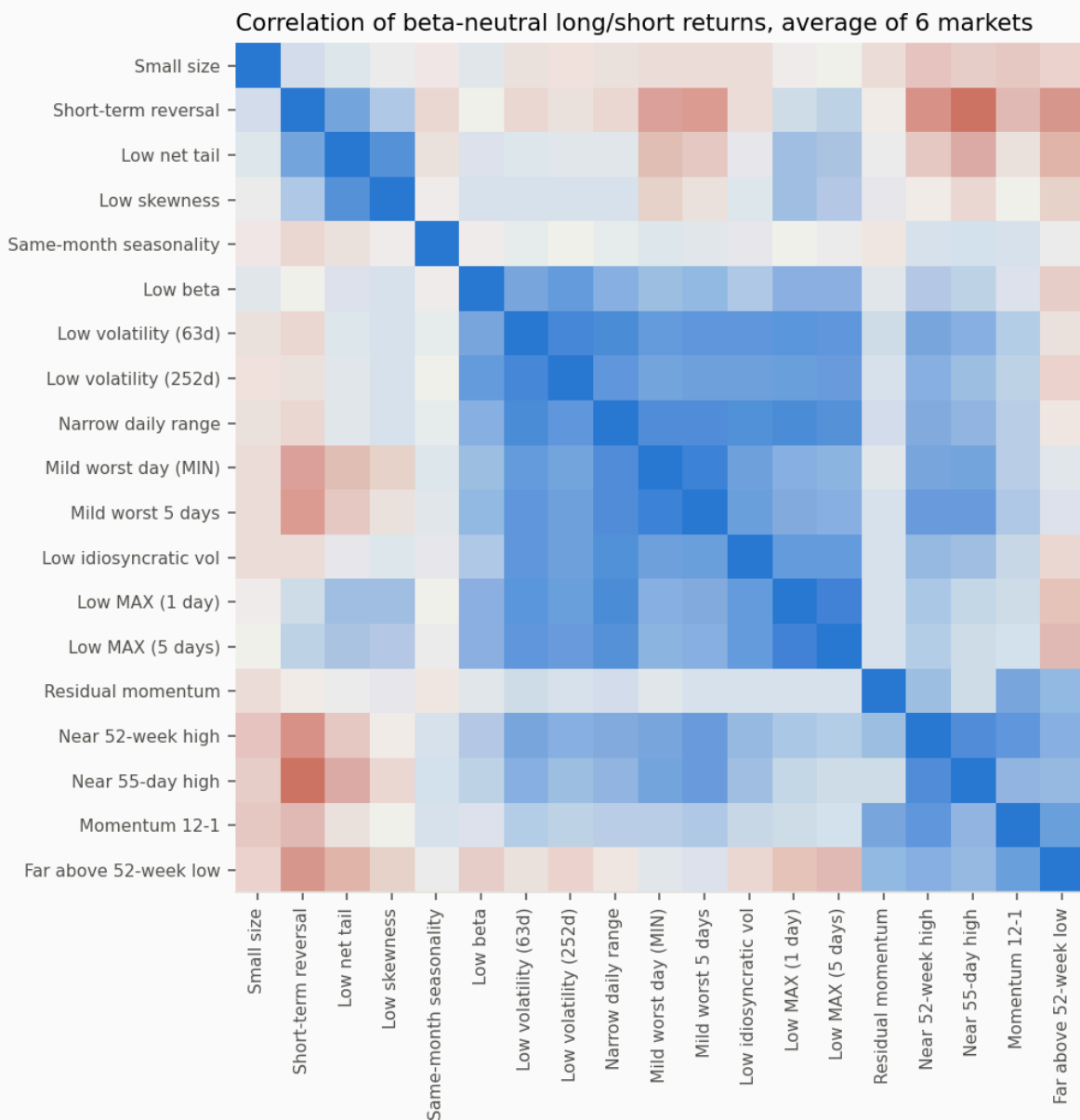
Averages are the equal-weight mean of the six market series; *t* is Newey–West on that mean. Effective markets = 36 / sum of the 6×6 correlation matrix. ✓ = discovery under Benjamini–Hochberg at 5% over 38 tests.

Reading it.

- The raw long/short numbers are mostly a beta story.** 9 of the 9 low-risk and tail-size signals have a negative raw average and 7 of 9 turn positive once the legs are beta-neutral and financed. Their raw books are short the market in a market that rose.
- Beta-neutral low-risk books need a lot of leverage.** To bring the low-beta leg up to a beta of one, the low-beta book borrows on average 0.8–2.9 units of cash per unit of capital (US 2.2, EU 2.3, UK 2.9, DK 0.8, SCANDI 1.1, World 2.4). Charged at the USD risk-free rate plus 50 bp, low beta still earns +10.0% a year (Sharpe 0.49, *t* 2.4, positive in 6 of 6), the strongest positive price signal, but it no longer clears the correction. The charge is conservative for EU and Denmark, where short rates were negative for 2015–21.
- After the correction: 1 discovery out of 38:** Same-month seasonality (beta-neutral) (*t* -3.4). The strongest positive candidates are low beta (*t* +2.4) and 12-1 momentum (*t* +2.0); neither clears the bar.
- Momentum holds in most markets in both books:** 12-1 momentum is positive raw in 5 and beta-neutral in 5 of six markets, residual momentum in 6 and 6. Positive raw in all six: Residual momentum, Far above 52-week low.
- Tail direction is not priced the way the lottery literature expects** in our large caps: low skewness and low net tail lose in most markets, raw and beta-neutral.

6. **Costs sort the families:** the within-month signals (MAX, MIN, skewness, reversal, seasonality) cost 3–4% a year; volatility, beta, momentum and size cost 0–2%.

3. Common ground among signals: two blocks



Clustering the beta-neutral long/short returns (average correlation over the six markets, average linkage, cut at correlation 0.5) gives:

Signals in the group	Size
Low MAX (1 day), Low MAX (5 days), Mild worst day (MIN), Mild worst 5 days, Narrow daily range, Low idiosyncratic vol, Low volatility (63d), Low volatility (252d), Low beta	9
Momentum 12-1, Near 52-week high, Far above 52-week low, Near 55-day high	4
Low net tail, Low skewness	2
Short-term reversal	1

Signals in the group	Size
Residual momentum	1
Same-month seasonality	1
Small size	1

The nineteen signals hold far fewer independent bets. Everything that measures the *size* of price moves — MAX, MIN, the daily range, idiosyncratic volatility, volatility and beta — is one block. Momentum, 52-week high, 52-week low distance and the breakout are a second block. Residual momentum, reversal, seasonality, tail direction and size stand alone.

Is the structure the same in every market? Adjusted Rand index between each market's own clustering and the pooled one: US 0.88, EU 0.87, UK 0.47, DK 0.70, SCANDI 0.46, World 0.97 (1 = identical, 0 = random). The two-block structure is almost identical in the US, EU and World and looser in the UK and Scandinavia, where small groups and fewer names make correlations noisier.

4. Common ground among segments: do markets agree on what works?

Spearman rank correlation between markets of the 19 beta-neutral Sharpe ratios (average against the other five markets):

Market	Average agreement with the other markets
US	0.58
EU	0.55
UK	0.40
DK	0.41
SCANDI	0.07
World	0.60

The same test on the 153 JKP factors (2013–2025 Sharpe ratios):

Segment	Average agreement with the other segments
US	0.55
UK	0.59
Denmark	0.24
World ex US	0.65
World	0.69

SCANDI is the odd one out. In the price battery SCANDI (0.07) and UK (0.40) agree least; in JKP it is Denmark (0.24). With ~19 stocks in OMXC25, ~62 in SCANDI and a JKP Danish universe dominated by a few large names, factor returns there are driven by single companies. The US, EU/World ex US and World agree most, partly because they overlap. The

median correlation of the same JKP factor across segments: US–World ex US 0.56, US–UK 0.43, US–Denmark 0.18; factor returns are mostly local, so a signal that works in several segments is genuine breadth, not one bet counted twice.

5. The published factors: 153 JKP factors since 2013

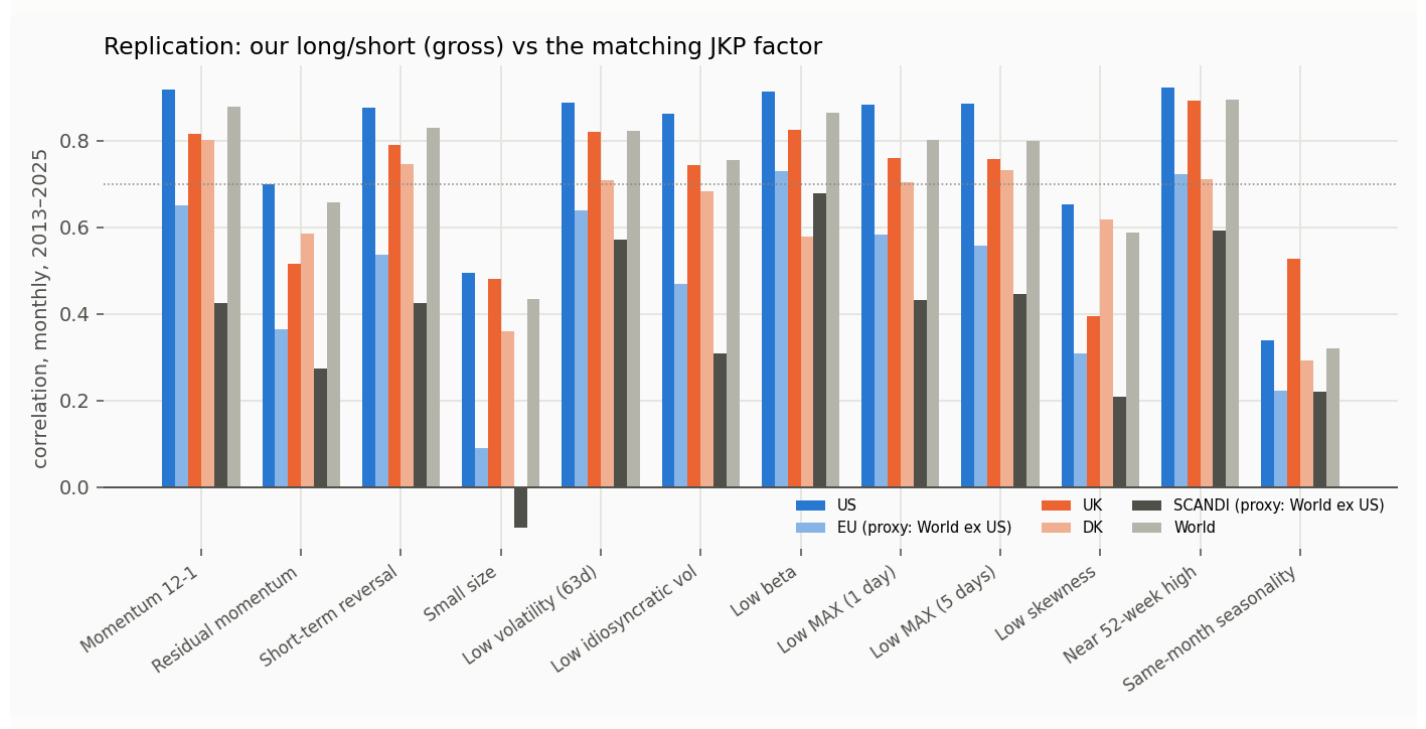
JKP's 13 themes (153 published factors, equal-weighted within theme): Sharpe ratio by segment											
Before 2013 (each segment's own history)						2013–2025					
Accruals	+0.76	+0.29	+0.35	+0.41	+0.75	Accruals	+0.03	+0.04	-0.34	+0.57	+0.30
Debt Issuance	+1.07	+0.65	+0.64	+0.26	+1.07	Debt Issuance	+0.43	+1.36	+1.03	+1.11	+0.89
Investment	+0.42	+0.35	+0.30	+0.45	+0.42	Investment	+0.12	+0.35	+0.41	+0.25	+0.20
Low Leverage	+0.01	-0.17	+0.14	-0.26	-0.01	Low Leverage	+0.10	-0.12	+0.15	-0.07	+0.04
Low Risk	+0.07	+0.23	+0.16	+0.43	+0.09	Low Risk	+0.07	+0.06	-0.11	+0.33	+0.18
Momentum	+0.28	+0.49	+0.52	+0.34	+0.29	Momentum	+0.42	+0.55	+0.89	+1.06	+0.78
Profit Growth	+0.44	+0.43	+0.41	+0.63	+0.45	Profit Growth	+0.34	+0.89	+0.33	+1.16	+0.81
Profitability	+0.29	+0.53	+0.34	+0.39	+0.29	Profitability	+0.31	+0.58	-0.29	+0.72	+0.54
Quality	+0.19	+0.65	+0.44	+0.39	+0.18	Quality	+0.46	+0.45	-0.03	+0.58	+0.57
Seasonality	+0.38	+1.02	+0.31	+0.45	+0.36	Seasonality	-0.08	-0.17	-0.16	+0.48	+0.20
Short-Term Reversal	+0.67	-0.40	-0.05	+0.24	+0.71	Short-Term Reversal	+0.10	-0.04	-0.25	+0.14	+0.12
Size	+0.27	-0.03	-0.31	+0.55	+0.31	Size	-0.15	+0.11	+0.55	+0.74	+0.19
Value	+0.29	+0.39	+0.29	+0.90	+0.36	Value	+0.15	+0.57	+0.21	+0.47	+0.30
	US	UK	Denmark	World ex US	World		US	UK	Denmark	World ex US	World

Pooled over the four disjoint segments (US, UK, Denmark, World ex US), **19 of 153 factors** are discoveries under Benjamini–Hochberg at 5% and 14 clear the Harvey–Liu–Zhu bar of $t > 3$. The strongest twelve:

Factor	Theme	Average return / yr	Sharpe	t	Positive in	Before 2013 (4-segment average)	BH
nfna_gr1a	Debt Issuance	+4.5%	1.34	+5.3	4 / 4	+3.3%	✓
fnl_gr1a	Debt Issuance	+4.3%	1.20	+5.1	4 / 4	+3.8%	✓
noa_at	Debt Issuance	+5.0%	1.07	+5.0	4 / 4	+4.1%	✓
resff3_12_1	Momentum	+5.3%	1.06	+4.2	4 / 4	+7.9%	✓
seas_1_1na	Momentum	+8.7%	1.02	+4.2	4 / 4	+3.9%	✓
rd_me	Size	+6.8%	1.00	+3.9	4 / 4	+3.9%	✓
ret_9_1	Momentum	+7.4%	0.84	+3.8	4 / 4	+6.1%	✓
saleq_su	Profit Growth	+5.6%	1.01	+3.8	4 / 4	+0.2%	✓
ret_6_1	Momentum	+5.7%	0.74	+3.7	4 / 4	+4.0%	✓
capx_gr1	Investment	+4.2%	0.87	+3.3	4 / 4	+3.2%	✓
niq_be_chg1	Profit Growth	+4.4%	0.93	+3.2	4 / 4	+4.1%	✓
ret_12_1	Momentum	+7.3%	0.83	+3.1	4 / 4	+8.1%	✓

Reading it. The winners since 2013 come from three themes: **momentum** (12-1, 9-1, 6-1, residual momentum, 52-week high, seasonality at lag 1), **debt issuance** (firms that shrink net financing, debt or net operating assets beat those that raise them) and **profit growth** (sales and earnings surprises and changes). Value, investment and accruals, stars of the pre-2013 literature, weakened sharply in the US; short-term reversal fell from a Sharpe of 0.67 to 0.10 there.

6. Replication check: does our engine match the published factors?



Our signal	JKP factor	US	EU (proxy)	UK	DK	SCANDI (proxy)	World
Momentum 12-1	ret_12_1	0.92	0.65	0.82	0.80	0.43	0.88
Residual momentum	resff3_12_1	0.70	0.36	0.52	0.59	0.27	0.66
Short-term reversal	ret_1_0	0.88	0.54	0.79	0.75	0.42	0.83
Small size	market_equity	0.49	0.09	0.48	0.36	-0.09	0.44
Low volatility (63d)	rvol_21d	0.89	0.64	0.82	0.71	0.57	0.82
Low idiosyncratic vol	ivol_capm_21d	0.86	0.47	0.74	0.68	0.31	0.76
Low beta	beta_60m	0.91	0.73	0.83	0.58	0.68	0.86
Low MAX (1 day)	rmax1_21d	0.88	0.58	0.76	0.70	0.43	0.80
Low MAX (5 days)	rmax5_21d	0.89	0.56	0.76	0.73	0.45	0.80
Low skewness	rskew_21d	0.65	0.31	0.39	0.62	0.21	0.59
Near 52-week high	prc_highprc_252d	0.92	0.72	0.89	0.71	0.59	0.89
Same-month seasonality	seas_1_1an	0.34	0.22	0.53	0.29	0.22	0.32

In the US, where the universes are closest (our top 500 vs JKP's capped value-weighted full market), **8 of 12 matched signals correlate 0.8 or more** with the published factor (Momentum 12-1, Short-term reversal, Low volatility (63d), Low idiosyncratic vol, Low beta, Low MAX (1 day), Low MAX (5 days), Near 52-week high). This closes the replication gap raised in FS00: the engine produces the known factors. The weaker matches have known reasons: our seasonality averages up to 13 years of same-month returns while JKP's `seas_1_1an` uses only last year's; our size sort is within the 500 largest (JKP spans micro caps). EU and Scandinavia are compared with World ex US, so their lower numbers measure the proxy, not the engine.

7. Where is the common ground?

Theme	Our price data: t (markets > 0)	JKP: t (segments > 0)	JKP US Sharpe before 2013 → since	JKP factors that are BH discoveries	Verdict
Momentum	+1.4 (5/6)	+4.4 (4/4)	+0.28 → +0.42	6 / 8	Positive almost everywhere, not significant
Low risk	+1.6 (5/6)	+0.2 (3/4)	+0.07 → +0.07	0 / 18	No common ground
Lottery & tails (size of tails)	+0.0 (3/6)	—	—	—	No common ground
Tail direction (skewness)	-2.7 (1/6)	—	—	—	Reliably negative
Short-term reversal	-1.9 (1/6)	-0.5 (2/4)	+0.67 → +0.10	0 / 6	No common ground · faded in the US
Seasonality	-3.4 (1/6)	-0.5 (1/4)	+0.38 → -0.08	1 / 12	Reliably negative · faded in the US
Size	+0.4 (2/6)	+1.1 (3/4)	+0.27 → -0.15	1 / 5	No common ground
Breakout	-0.1 (3/6)	—	—	—	No common ground
Value	—	+1.0 (4/4)	+0.29 → +0.15	1 / 18	Positive almost everywhere, not significant
Quality	—	+1.2 (3/4)	+0.19 → +0.46	0 / 17	Positive almost everywhere, not significant
Profitability	—	+0.9 (3/4)	+0.29 → +0.31	0 / 11	Positive almost everywhere, not significant
Profit growth	—	+2.5 (4/4)	+0.44 → +0.34	4 / 12	Common ground (significant; JKP only)
Investment	—	+1.2 (4/4)	+0.42 → +0.12	1 / 22	Positive almost everywhere, not significant · faded in the US
Debt issuance	—	+6.2 (4/4)	+1.07 → +0.43	4 / 7	Common ground (significant; JKP only)
Accruals	—	-0.4 (3/4)	+0.76 → +0.03	1 / 6	No common ground · faded in the US
Low leverage	—	+0.2 (2/4)	+0.01 → +0.10	0 / 11	No common ground

Price t: beta-neutral composite of the theme's signals, average of 6 markets. JKP t: equal-weight theme portfolio, average of 4 disjoint segments, 2013–2025. "Common ground" = positive in every segment of every source that covers the theme, and pooled $t > 2$ in at least one source. "Faded in the US" = JKP US Sharpe above 0.3 before 2013 and below 0.15 since.

Common ground: Profit growth, Debt issuance. **Positive almost everywhere, not significant:** Momentum, Value, Quality, Profitability, Investment. **Works pooled, not everywhere:** none. **Reliably negative:** Tail direction (skewness), Seasonality. **Faded in the US:** Short-term reversal, Seasonality, Investment, Accruals.

Momentum is the strongest theme across both sources: positive in 5 of six of our markets and all 4 JKP segments; significant in JKP ($t +4.4$) but not in our large caps once short legs pay borrow fees ($t +1.4$); it misses the common-ground label because SCANDI is negative. **Low risk** is positive in 5 of six of our markets after financing ($t 1.6$) but only 3 of 4 JKP segments ($t 0.2$); JKP's low-risk factors are value-weighted and not beta-neutral, so they carry the short-market drag that our beta-neutral books remove. **Debt issuance** ($t 6.2$) and **profit growth** ($t 2.5$) are the only themes positive in every segment with a significant pooled return, and both are fundamental themes to add first. Value, quality, profitability and investment were positive in most JKP segments but not significant since 2013.

8. What this means for the multifactor model

The battery gives a pre-selection that is transparent and grounded in more than one segment:

1. **Carry forward:** low beta (beta-neutral, financed) as the low-risk representative, and 12-1 momentum, residual momentum and the 52-week high from the momentum block; the other signals in each block are near-duplicates.
2. **Add from fundamentals next** (US stock level from Sharadar; JKP factor level elsewhere): debt issuance and profit growth first, then quality and value as diversifiers even though they were weak since 2013.
3. **Drop:** same-month seasonality, short-term reversal and tail direction (skewness, net tail) in large caps; the lottery/tail-size signals add nothing beyond low beta and volatility.
4. **Treat Denmark as a robustness market, not a selection market:** its signal ranking agrees least with the others.
5. The next factsheet applies the planned filters (net return, Sharpe, turnover per segment), rolling 3-year correlations and the dynamic weighting to this shortlist, walk-forward.

9. Caveats

- **Different construction.** Our books are equal-weighted, net, local currency (World in USD), large caps; JKP's are capped value-weighted, gross, USD, broad market. Agreement across the two is therefore stronger evidence than either alone, but numbers are not comparable one-to-one.
- **Proxies.** EU and Scandinavia have no JKP data here; World ex US stands in. World overlaps US, UK and EU in both batteries; the pooled tests use disjoint segments for JKP and report the effective number of markets for the price battery.
- **Periods.** Price battery Feb 2013 – Aug 2026; JKP Jan 2013 – Dec 2025. JKP's "before 2013" history starts in the 1920s for the US and in 1987 for the UK, Denmark and World ex US.
- **Multiple testing.** 38 price tests and 153 JKP factors are corrected with Benjamini–Hochberg; the earlier factsheets' fix ladders are not in this count (see the FS00 gap on a programme-level ledger).

- **Three directions are conventions** (MIN, range, 52-week low distance); see §1.
- **Costs.** 10 bp per side, a size-tiered borrow fee on every short leg and financing of net long cash (USD risk-free + 50 bp) are charged; market impact is not (FS00).

10. References

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11. Reproduce

code/battery.py (price battery → results/battery_price.json/.pkl), code/jkp_battery.py (JKP battery → results/battery_jkp_*.csv), code/financing.py (financing charge for the beta-neutral books), code/common_ground.py (pooled tests, clusters, agreement, replication, themes, figures → results/common_ground.json), code/build_fs13.py (this factsheet). JKP monthly factor files come from Project2's reporting/style_cache.